

# Developer Economics & Language Trends (Stack Overflow 2015–2024)

## Course:

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## Notebook Description

This notebook explores the relationship between **developer-reported programming language usage**, **average salaries**, and **experience levels** using data from the **Stack Overflow Developer Survey (2015–2024)**.

While a core focus is the **correlation between language popularity and salary** (using **Pearson** and **Spearman** methods), the analysis extends further:

We investigate salary patterns by **country**, **language**, and **experience**, compute **trend scores**, and ultimately derive a **composite "Future Score"** that identifies promising language–experience–country combinations.

Data is processed using **Apache Spark**, visualized with **Matplotlib**, and normalized using **MinMaxScaler** to enable fair comparisons across diverse indicators.

## Goals of this Notebook

- Load and clean Stack Overflow survey data from 2015 to 2024.
- Measure language popularity by developer count per year.
- Analyze salary distributions by:
  - Programming language
  - Country
  - Developer experience
- Identify top-paying languages and regions.
- Compute language **trend scores** based on usage growth.
- Evaluate **correlation between language popularity and salary** (Pearson + Spearman).
- Derive a **Future Score** combining salary, popularity, trend, and regional factors.
- Highlight the best developer strategy by identifying top language–experience–country profiles.

## Analytical Dimensions

## 1. Popularity vs. Salary Correlation

- Pearson: linear relationships
- Spearman: rank-based monotonic relationships

## 2. Salary Analytics

- Per language
- Per country
- By experience level

## 3. Composite Analysis

- Combines trend, salary, experience, and geography
- Identifies top career-relevant combinations for developers

# Import Dependencies

Load essential libraries for Spark processing, data visualization, JSON handling, and logging.

```
In [1]: from pyspark.sql import SparkSession
from pyspark.sql.types import StructType, StructField, StringType, IntegerType,
from pyspark.sql.functions import split, explode, col, avg, count, trim
from pyspark.ml.feature import VectorAssembler
from pyspark.ml.stat import Correlation
import matplotlib.pyplot as plt
import pandas as pd
from sklearn.preprocessing import MinMaxScaler
from itertools import product
import logging
import pyspark.sql.functions as F
import json
logging.basicConfig(
    level=logging.INFO,
    format="%(asctime)s - %(levelname)s - %(message)s"
)

logger = logging.getLogger("KafkaProducer")
```

# Initialize Spark Session for S3-Based Analysis

Create a Spark session connected to a standalone cluster and MinIO-based S3 storage for reading Stack Overflow data.

```
In [2]: spark = SparkSession.builder \
    .appName("StackoverflowAnalysis") \
    .master("local[*]") \
    .config("spark.hadoop.fs.s3a.endpoint", "http://172.29.16.105:9000") \
    .config("spark.hadoop.fs.s3a.access.key", "bdenggroup4") \
    .config("spark.hadoop.fs.s3a.secret.key", "bdenggroup4") \
    .config("spark.hadoop.fs.s3a.path.style.access", "true") \
    .config("spark.hadoop.fs.s3a.impl", "org.apache.hadoop.fs.s3a.S3AFileSystem") \
    .getOrCreate()
```

```
spark.sparkContext.setLogLevel("ERROR")
```

Setting default log level to "WARN".

To adjust logging level use `sc.setLogLevel(newLevel)`. For SparkR, use `setLogLevel(newLevel)`.

25/06/23 11:31:48 WARN NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable

25/06/23 11:31:49 WARN Utils: Service 'SparkUI' could not bind on port 4040. Attempting port 4041.

25/06/23 11:31:49 WARN Utils: Service 'SparkUI' could not bind on port 4041. Attempting port 4042.

25/06/23 11:31:49 WARN Utils: Service 'SparkUI' could not bind on port 4042. Attempting port 4043.

25/06/23 11:31:49 WARN Utils: Service 'SparkUI' could not bind on port 4043. Attempting port 4044.

25/06/23 11:31:49 WARN Utils: Service 'SparkUI' could not bind on port 4044. Attempting port 4045.

25/06/23 11:31:49 WARN Utils: Service 'SparkUI' could not bind on port 4045. Attempting port 4046.

25/06/23 11:31:49 WARN Utils: Service 'SparkUI' could not bind on port 4046. Attempting port 4047.

## Load and Inspect Stack Overflow Survey Data

Read cleaned survey CSV files from S3 with a predefined schema and display a sample along with the data structure.

```
In [3]: schema = StructType() \
        .add("ConvertedCompYearly", DoubleType()) \
        .add("LanguageHaveWorkedWith", StringType()) \
        .add("MainBranch", StringType()) \
        .add("Country", StringType()) \
        .add("YearsCode", StringType()) \
        .add("LanguageList", StringType()) \
        .add("Year", StringType())

data_path = "s3a://bdenggroup4/survey_results_public_cleaned_*.csv"
df = spark.read.option("header", True).schema(schema).csv(data_path)

df.printSchema()
df.show(5, truncate=False)
```

```

root
|-- ConvertedCompYearly: double (nullable = true)
|-- LanguageHaveWorkedWith: string (nullable = true)
|-- MainBranch: string (nullable = true)
|-- Country: string (nullable = true)
|-- YearsCode: string (nullable = true)
|-- LanguageList: string (nullable = true)
|-- Year: string (nullable = true)

+-----+-----+-----+-----+
+-----+-----+-----+-----+
+-----+-----+
|ConvertedCompYearly|LanguageHaveWorkedWith|MainBranch
|Country          |YearsCode|LanguageList
|Year|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
+-----+-----+
|61000.0          |C; C++; C#; Python|I am a develo
per by profession|United States|3.0      |['C', 'C++', 'C#', 'Python']
|2019|
|95179.0          |C#; JavaScript; Python; Ruby; Rust; TypeScript|I am a develo
per by profession|New Zealand  |12.0    |['C#', 'JavaScript', 'Python', 'Ruby',
'Rust', 'TypeScript']|2019|
|13293.0          |C#; Go; JavaScript; Python; R|I am a develo
per by profession|India        |12.0    |['C#', 'Go', 'JavaScript', 'Python',
'R']|2019|
|90000.0          |JavaScript; PHP; TypeScript|I am a develo
per by profession|United States|17.0    |['JavaScript', 'PHP', 'TypeScript']
|2019|
|57060.0          |C++|I am a develo
per by profession|Germany      |13.0    |['C++']
|2019|
+-----+-----+-----+-----+
+-----+-----+-----+-----+
+-----+-----+
only showing top 5 rows

```

## Clean and Prepare Language-Salary Data

Drop rows for safety with missing salary or language info, split multi-language responses, and create a flattened view of language-year-salary entries for analysis.

```

In [4]: df_clean = df.dropna(subset=["ConvertedCompYearly", "LanguageHaveWorkedWith", "Y

df_exploded = df_clean.withColumn("Language", explode(split(col("LanguageHaveWor
                .withColumn("Language", trim(col("Language")))) \
                .select("Year", "Language", "ConvertedCompYearly", "YearsC

df_exploded.createOrReplaceTempView("language_salary_raw")

```

## Create Yearly Language Count View

Aggregate the number of respondents per language and year to analyze popularity trends over time.

```
In [5]: spark.sql("""
        CREATE OR REPLACE TEMP VIEW language_yearly_counts AS
        SELECT
            Year AS year,
            Language AS language,
            COUNT(*) AS repo_count
        FROM language_salary_raw
        GROUP BY Year, Language
        """)
```

```
Out[5]: DataFrame[]
```

## Identify Top 5 Programming Languages Per Year by Survey Popularity

Use a SQL window function ( `ROW_NUMBER()` ) to rank programming languages by the number of developer survey mentions per year.

Select the top 5 most popular languages for each year to analyze yearly shifts in developer preferences based on reported usage. ``

```
In [6]: top_count_lang_per_year = spark.sql("""
        SELECT year, language, repo_count
        FROM (
            SELECT year,
                   language,
                   repo_count,
                   ROW_NUMBER() OVER (PARTITION BY year ORDER BY repo_count DESC) AS
            FROM language_yearly_counts
        )
        WHERE rank <= 5
        ORDER BY year
        """)

top_langs_pd = top_count_lang_per_year.toPandas()

top_langs_pd = top_langs_pd.dropna(subset=["year", "repo_count", "language"])
top_langs_pd["year"] = top_langs_pd["year"].astype(int)
top_langs_pd = top_langs_pd.sort_values(by=["year"])
```

## Visualize Top 5 Programming Languages Per Year (Stacked Bar Chart)

Create a stacked bar chart showing the five most popular programming languages per year, based on developer counts from the Stack Overflow survey.

The chart highlights relative language usage over time and includes annotations for the total developer count per year and the single most-used

```
In [7]: top_langs_pd = top_count_lang_per_year.toPandas().sort_values(["year", "repo_cou

pivot_df = top_langs_pd.pivot(index="year", columns="language", values="repo_cou

fig, ax = plt.subplots(figsize=(12, 7))
pivot_df.plot(kind="bar", stacked=True, colormap="tab20", alpha=0.9, ax=ax)

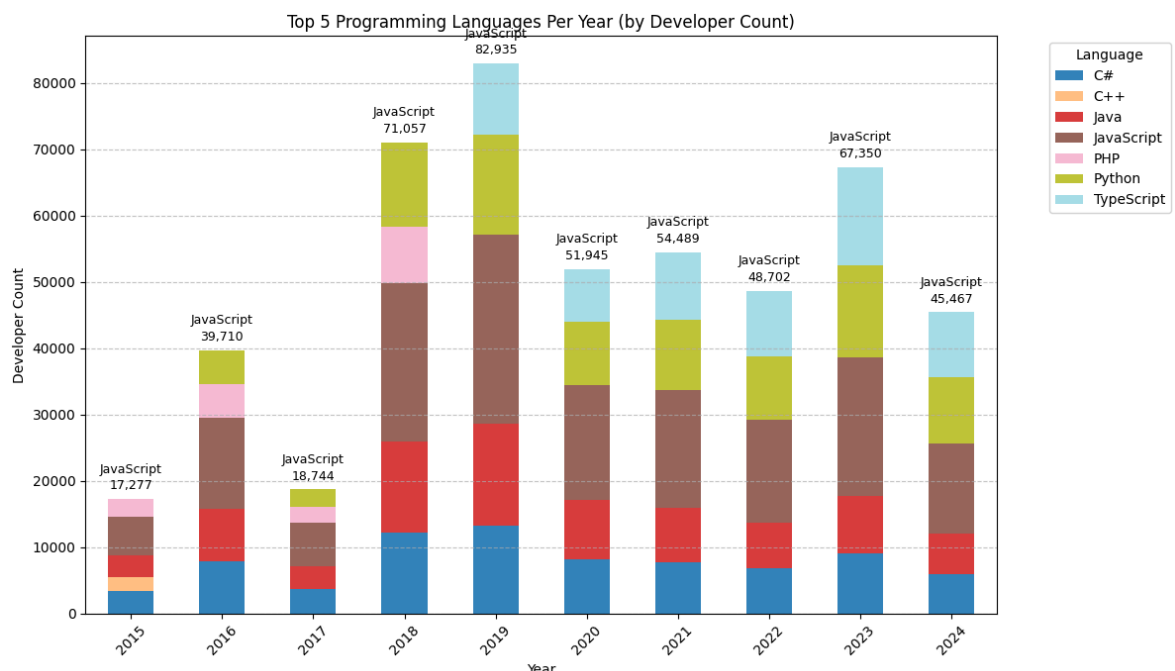
top1_langs = top_langs_pd.groupby("year").first().reset_index()

for i, year in enumerate(pivot_df.index):
    count = int(pivot_df.loc[year].sum())
    top_lang = top1_langs[top1_langs["year"] == year]["language"].values[0]

    ax.text(i, count + 1000, f"{count:}", ha="center", va="bottom", fontsize=9,
    ax.text(i, count + 3500, f"{top_lang}", ha="center", va="bottom", fontsize=9

ax.set_title("Top 5 Programming Languages Per Year (by Developer Count)")
ax.set_xlabel("Year")
ax.set_ylabel("Developer Count")
ax.set_xticklabels(pivot_df.index, rotation=45)
ax.grid(True, axis="y", linestyle="--", alpha=0.7)
ax.legend(title="Language", bbox_to_anchor=(1.05, 1), loc="upper left")

plt.tight_layout()
plt.show()
```



## Create View of Average Salary per Language and Year

Compute average yearly salary for each language, filtering out entries with zero salary and languages with low response counts.

```
In [8]: spark.sql("""
CREATE OR REPLACE TEMP VIEW language_yearly_salary AS
```

```

SELECT
    Year AS year,
    Language AS language,
    COUNT(*) AS count,
    AVG(ConvertedCompYearly) AS avg_salary
FROM language_salary_raw
WHERE ConvertedCompYearly > 0
GROUP BY Year, Language
HAVING COUNT(*) > 50

"""

```

Out[8]: DataFrame[]

## Identify Top Paying Language per Year

Use a window function to rank languages by average salary per year and select the highest-paying language annually.

```

In [9]: top_salary_lang_per_year = spark.sql("""
        SELECT year, language, avg_salary
        FROM (
            SELECT
                year,
                language,
                avg_salary,
                ROW_NUMBER() OVER (PARTITION BY year ORDER BY avg_salary DESC) AS rank
            FROM language_yearly_salary
        )
        WHERE rank = 1
        ORDER BY year
    """)

top_salary_pd = top_salary_lang_per_year.toPandas()
top_salary_pd = top_salary_pd.dropna()
top_salary_pd["year"] = top_salary_pd["year"].astype(int)
top_salary_pd = top_salary_pd.sort_values(by="year")

```

## Visualize Highest-Paying Language Per Year

Generate a line chart to show which programming language had the highest average salary each year according to survey data.

```

In [10]: pivot_salary_df = top_salary_pd.pivot(index="year", columns="language", values="avg_salary")

ax = pivot_salary_df.plot(kind="bar", stacked=True, figsize=(12, 7), colormap="tab10")

plt.title("Top Paying Programming Languages Per Year (Stacked Bar - Avg. Salary)")
plt.xlabel("Year")
plt.ylabel("Average Salary (USD)")
plt.xticks(rotation=45)
plt.grid(True, axis='y')
plt.legend(title="Language", bbox_to_anchor=(1.05, 1), loc="upper left")
plt.tight_layout()

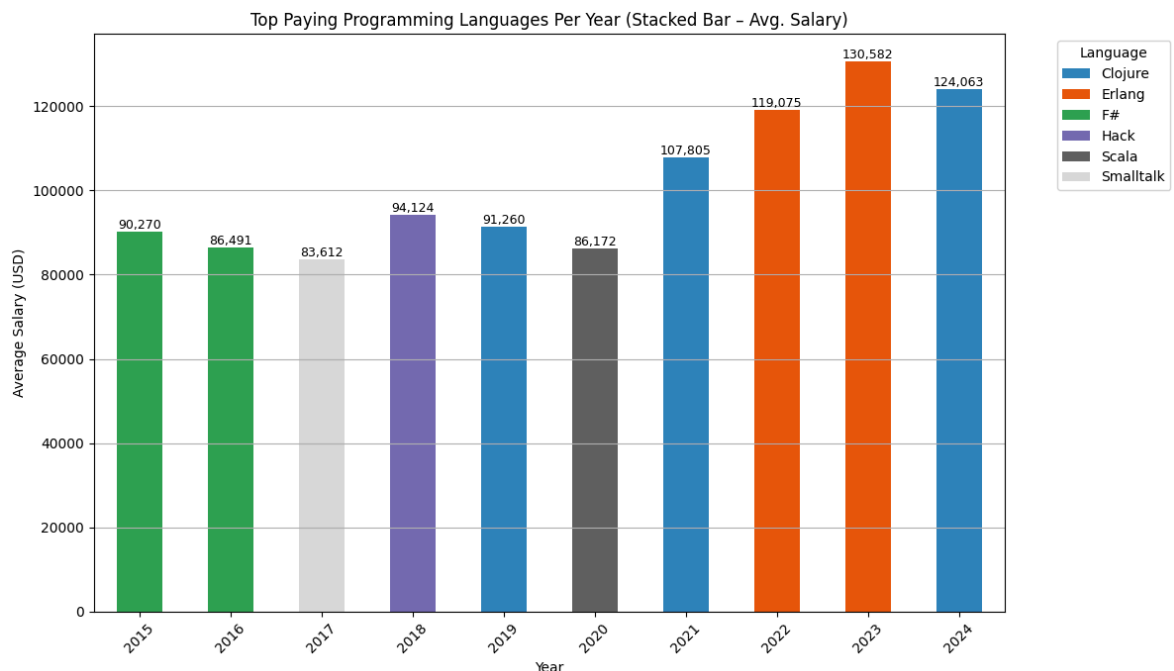
```

```

for p in ax.patches:
    height = p.get_height()
    if height > 0:
        ax.annotate(f'{int(height):,}',
                    (p.get_x() + p.get_width() / 2, p.get_y() + height),
                    ha='center', va='bottom', fontsize=9)

plt.show()

```



## Prepare Country-Level Salary Data

Select valid rows with Country and salary, excluding nulls and non-positive sal

```

In [11]: df_country_salary = df.select("Country", "ConvertedCompYearly") \
        .dropna(subset=["Country", "ConvertedCompYearly"]) \
        .filter(col("ConvertedCompYearly") > 0)

```

## Compute Average Salary by Country

Group responses by country and calculate average salary and response count, keeping only countries with at least 100 entries.

```

In [12]: df_avg_salary_country = df_country_salary.groupBy("Country") \
        .agg(
            F.avg("ConvertedCompYearly").alias("avg_salary"),
            F.count("*").alias("count")
        ) \
        .filter("count >= 100") \
        .orderBy(F.col("avg_salary").desc())

```

## Visualize Top Paying Countries

Convert salary data to Pandas, extract top 20 countries by average salary, and plot them as a horizontal bar chart.



```
In [13]: df_avg_salary_pd = df_avg_salary_country.toPandas()
df_avg_salary_pd["avg_salary"] = df_avg_salary_pd["avg_salary"].round(2)

top_n = 20
top_countries = df_avg_salary_pd.head(top_n)

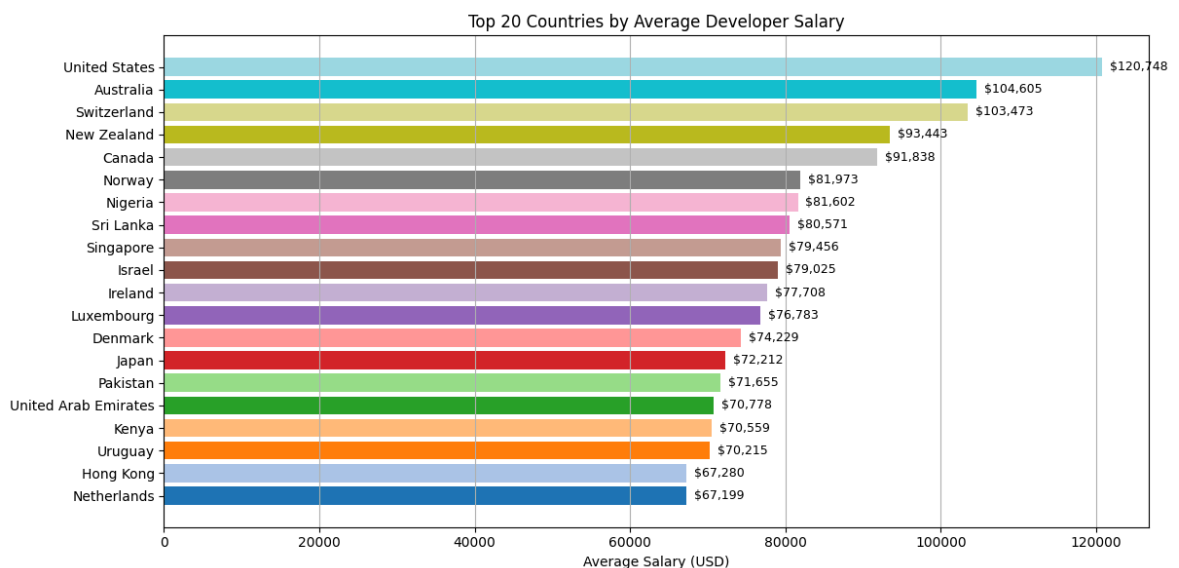
plt.figure(figsize=(12, 6))
colors = plt.cm.tab20.colors[:top_n]

bars = plt.barh(
    top_countries["Country"][:-1],
    top_countries["avg_salary"][:-1],
    color=colors
)

plt.xlabel("Average Salary (USD)")
plt.title(f"Top {top_n} Countries by Average Developer Salary")
plt.tight_layout()
plt.grid(axis="x")

for bar in bars:
    width = bar.get_width()
    plt.text(width + 1000, # Abstand rechts neben Balken
             bar.get_y() + bar.get_height() / 2,
             f"${int(width):,}",
             va='center', fontsize=9)

plt.show()
```



## Calculate Average Salary Per Language

Group by programming language, compute average salary and count, filter out small samples, and sort by salary descending.

```
In [14]: df_avg_salary_per_lang = df_exploded.groupby("Language") \
    .agg(
        F.avg("ConvertedCompYearly").alias("avg_salary"),
        F.count("*").alias("count")
    ) \
```

```
.filter("count >= 50") \
.orderBy(col("avg_salary").desc())
```

## Plot Top Programming Languages by Average Salary

Convert results to Pandas, extract top 20 languages by salary, and visualize with a horizontal bar chart including salary labels.

```
In [15]: df_avg_salary_pd = df_avg_salary_per_lang.toPandas()
df_avg_salary_pd["avg_salary"] = df_avg_salary_pd["avg_salary"].round(2)

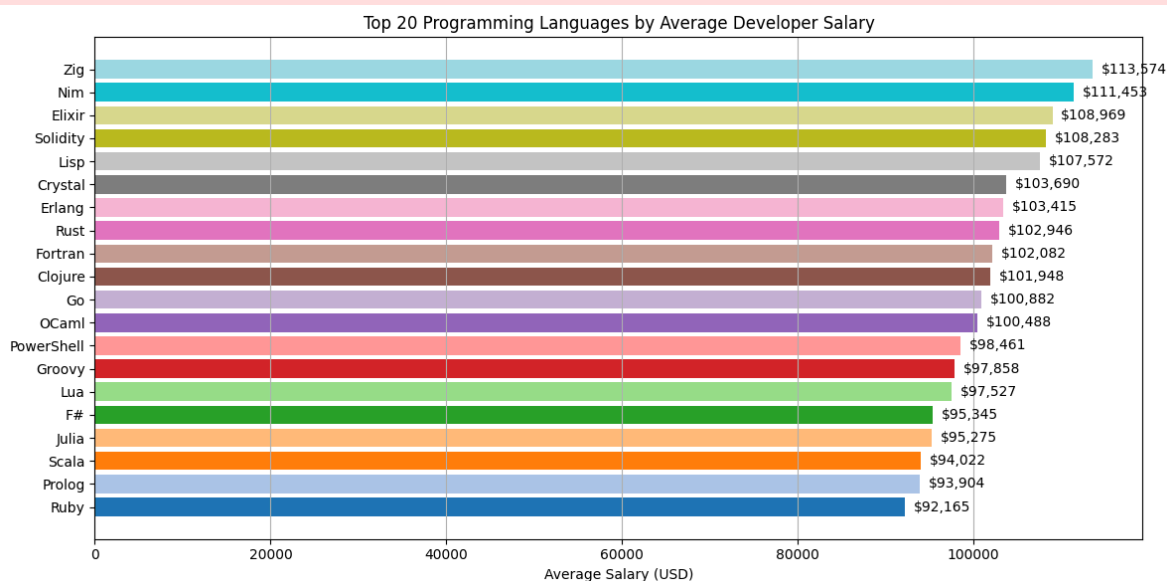
top_n = 20
top_languages = df_avg_salary_pd.head(top_n)

plt.figure(figsize=(12, 6))
colors = plt.cm.tab20.colors[:top_n]

bars = plt.barh(
    top_languages["Language"][::-1],
    top_languages["avg_salary"][::-1],
    color=colors
)

for bar, value in zip(bars, top_languages["avg_salary"][::-1]):
    plt.text(
        bar.get_width() + 1000,
        bar.get_y() + bar.get_height() / 2,
        f"${value:,.0f}",
        va='center'
    )

plt.xlabel("Average Salary (USD)")
plt.title(f"Top {top_n} Programming Languages by Average Developer Salary")
plt.tight_layout()
plt.grid(axis="x")
plt.show()
```



## Combine Top and Common Programming Languages

Select top 20 languages by average salary and append popular languages, then remove duplicates and sort by salary.

```
In [16]: common_languages = {
    "Python", "JavaScript", "Java", "C#", "C++", "C", "TypeScript", "Go", "PHP",
}

top_n = 20
top_languages = df_avg_salary_pd.sort_values(by="avg_salary", ascending=False).h
common_lang_rows = df_avg_salary_pd[df_avg_salary_pd["Language"].isin(common_lang
combined = pd.concat([top_languages, common_lang_rows]).drop_duplicates(subset=[
combined = combined.sort_values(by="avg_salary", ascending=False)
```

## Batch Plot Programming Languages by Salary

Split language-salary data into chunks of N entries, highlight common languages in orange, and generate one horizontal bar chart per batch.

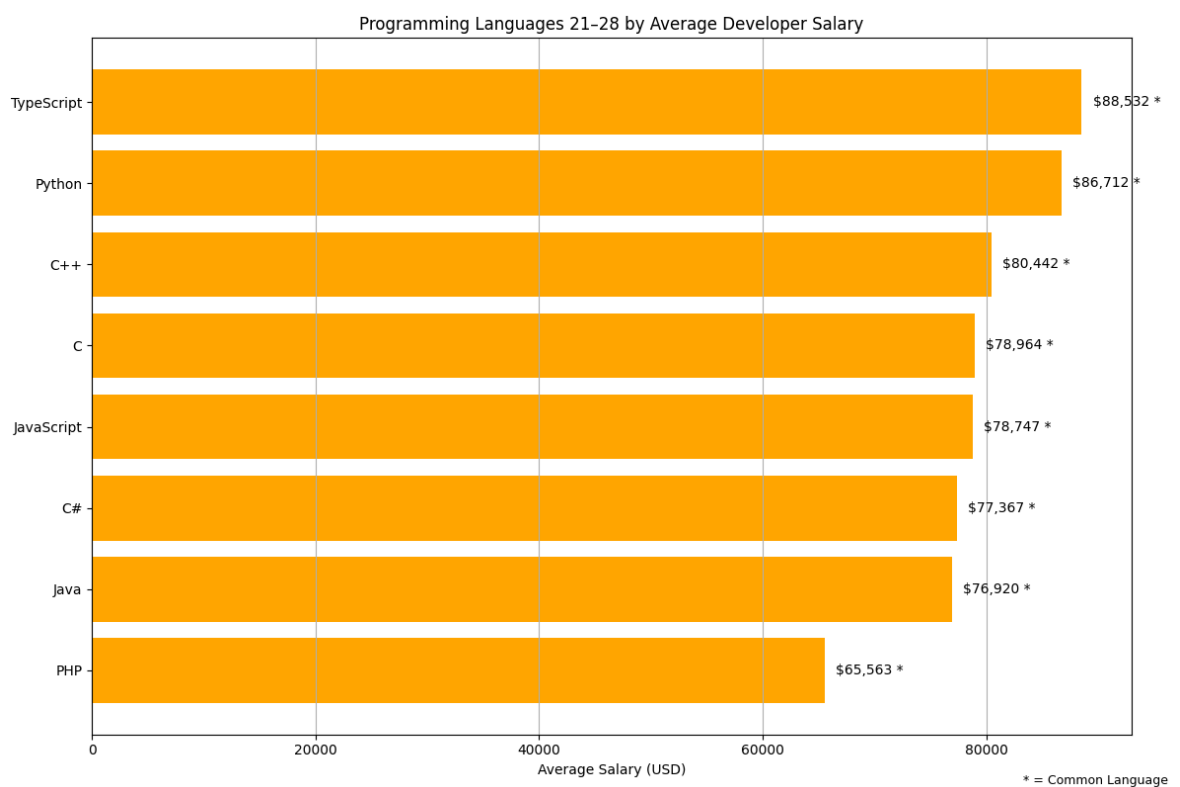
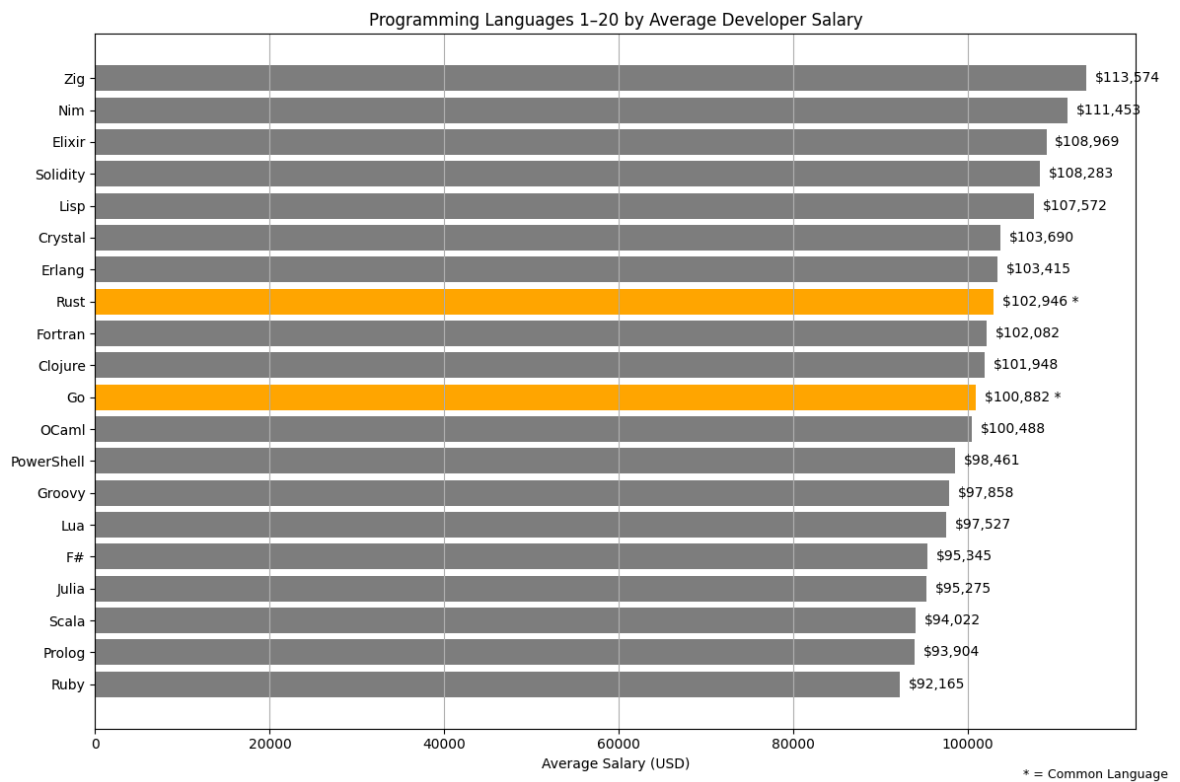
```
In [17]: batch_size = 20
combined_sorted = combined.sort_values(by="avg_salary", ascending=False).reset_i

for i in range(0, len(combined_sorted), batch_size):
    batch = combined_sorted.iloc[i:i+batch_size]
    colors = ["orange" if lang in common_languages else "gray" for lang in batch

    plt.figure(figsize=(12, 8))
    bars = plt.barh(
        batch["Language"][:-1],
        batch["avg_salary"][:-1],
        color=colors
    )

    for bar, lang, value in zip(bars, batch["Language"][:-1], batch["avg_salary
        label = f"${value:,.0f}"
        if lang in common_languages:
            label += " *"
        plt.text(bar.get_width() + 1000, bar.get_y() + bar.get_height()/2, label

    plt.xlabel("Average Salary (USD)")
    plt.title(f"Programming Languages {i+1}-{i+len(batch)} by Average Developer
    plt.grid(axis="x")
    plt.tight_layout()
    plt.figtext(0.99, 0.01, "* = Common Language", ha="right", fontsize=9)
    plt.show()
```



## Top 10 Experience Levels by Average Salary

Compute the average salary per years of coding experience using PySpark.

Filter out invalid or low-frequency entries, and display the top 10 experience levels sorted by average salary (only levels with  $\geq 50$  responses are considered).

```
In [18]: df_exp_salary = df.select("YearsCode", "ConvertedCompYearly") \
        .dropna(subset=["YearsCode", "ConvertedCompYearly"]) \
        .filter(col("ConvertedCompYearly") > 0)
```

```

df_avg_salary_by_exp = df_exp_salary.groupby("YearsCode") \
    .agg(
        F.avg("ConvertedCompYearly").alias("avg_salary"),
        F.count("*").alias("count")
    ) \
    .filter("count >= 50") \
    .orderBy(col("avg_salary").desc())

df_top_exp_pd = df_avg_salary_by_exp.toPandas()
df_top_exp_pd["avg_salary"] = df_top_exp_pd["avg_salary"].round(2)

print(" Top 10 experience levels by average salary:")
print(df_top_exp_pd.head(10))

```

Top 10 experience levels by average salary:

|   | YearsCode | avg_salary | count |
|---|-----------|------------|-------|
| 0 | 28.0      | 118203.63  | 1717  |
| 1 | 27.0      | 113897.92  | 1586  |
| 2 | 30.0      | 113802.77  | 4693  |
| 3 | 26.0      | 113627.98  | 1704  |
| 4 | 25.0      | 112049.06  | 5832  |
| 5 | 24.0      | 111133.89  | 2295  |
| 6 | 23.0      | 110900.93  | 2576  |
| 7 | 29.0      | 110595.96  | 759   |
| 8 | 22.0      | 107766.02  | 3995  |
| 9 | 21.0      | 104768.89  | 2224  |

## Line Plot: Average Developer Salary by Years of Experience

Convert and clean experience-salary data, sort by years, and visualize the average salary trend across different levels of coding experience using a line chart.

```

In [19]: df_avg_salary_exp_pd = df_avg_salary_by_exp.toPandas()

df_avg_salary_exp_pd = df_avg_salary_exp_pd[
    pd.to_numeric(df_avg_salary_exp_pd["YearsCode"], errors="coerce").notna()
]
df_avg_salary_exp_pd["YearsCode"] = df_avg_salary_exp_pd["YearsCode"].astype(float)

df_avg_salary_exp_pd = df_avg_salary_exp_pd.sort_values(by="YearsCode")
df_avg_salary_exp_pd["avg_salary"] = df_avg_salary_exp_pd["avg_salary"].round(2)

plt.figure(figsize=(12, 6))
plt.plot(
    df_avg_salary_exp_pd["YearsCode"],
    df_avg_salary_exp_pd["avg_salary"],
    marker="o"
)

plt.title("Average Developer Salary by Years of Experience")
plt.xlabel("Years of Coding Experience")
plt.ylabel("Average Salary (USD)")
plt.grid(True)
plt.tight_layout()
plt.show()

```



## Create Combined View of Popularity and Salary for Correlation

Build a temporary view that links each language's popularity (count) with its average salary per year for correlation analysis.

```
In [20]: spark.sql("""
CREATE OR REPLACE TEMP VIEW language_popularity_salary AS
SELECT
    Year AS year,
    Language AS language,
    COUNT(*) AS popularity,
    AVG(ConvertedCompYearly) AS avg_salary
FROM language_salary_raw
WHERE ConvertedCompYearly > 0
GROUP BY Year, Language
HAVING COUNT(*) > 50
""")
```

Out[20]: DataFrame[]

```
In [21]: years = list(range(2015, 2025))

for y in years:
    df_debug = spark.sql(f"""
        SELECT year, language, popularity, avg_salary
        FROM language_popularity_salary
        WHERE year = {y}
        """)

    df_debug_pd = df_debug.toPandas()

    if not df_debug_pd.empty:
        top5 = df_debug_pd.sort_values(by="popularity", ascending=False).head(5)
        print(f"\n Top 5 Languages for Year {y} sorted by popularity:")
        print(top5.to_string(index=False))
    else:
        print(f"\n No Data for Year: {y}")
```

Top 5 Languages for Year 2015 sorted by popularity:

| year | language   | popularity | avg_salary   |
|------|------------|------------|--------------|
| 2015 | JavaScript | 5780       | 56103.806228 |
| 2015 | C#         | 3371       | 60602.195194 |
| 2015 | Java       | 3327       | 54598.737601 |
| 2015 | PHP        | 2660       | 45105.263158 |
| 2015 | C++        | 2139       | 62519.869098 |

Top 5 Languages for Year 2016 sorted by popularity:

| year | language   | popularity | avg_salary   |
|------|------------|------------|--------------|
| 2016 | JavaScript | 13758      | 62430.585841 |
| 2016 | C#         | 7904       | 66344.888664 |
| 2016 | Java       | 7832       | 64682.073544 |
| 2016 | PHP        | 5169       | 52794.544399 |
| 2016 | Python     | 5047       | 71200.713295 |

Top 5 Languages for Year 2017 sorted by popularity:

| year | language   | popularity | avg_salary   |
|------|------------|------------|--------------|
| 2017 | JavaScript | 6553       | 64552.824508 |
| 2017 | C#         | 3764       | 66025.193624 |
| 2017 | Java       | 3442       | 64272.139831 |
| 2017 | Python     | 2680       | 69621.890784 |
| 2017 | PHP        | 2305       | 54810.748677 |

Top 5 Languages for Year 2018 sorted by popularity:

| year | language   | popularity | avg_salary   |
|------|------------|------------|--------------|
| 2018 | JavaScript | 23870      | 69685.460494 |
| 2018 | Java       | 13685      | 68653.451589 |
| 2018 | Python     | 12678      | 74444.695378 |
| 2018 | C#         | 12253      | 69598.551294 |
| 2018 | PHP        | 8571       | 59426.551861 |

Top 5 Languages for Year 2019 sorted by popularity:

| year | language   | popularity | avg_salary   |
|------|------------|------------|--------------|
| 2019 | JavaScript | 28635      | 67986.033316 |
| 2019 | Java       | 15252      | 67106.435484 |
| 2019 | Python     | 14983      | 73622.176333 |
| 2019 | C#         | 13323      | 66910.721834 |
| 2019 | TypeScript | 10742      | 70410.354589 |

Top 5 Languages for Year 2020 sorted by popularity:

| year | language   | popularity | avg_salary   |
|------|------------|------------|--------------|
| 2020 | JavaScript | 17314      | 66220.012418 |
| 2020 | Python     | 9604       | 72029.181799 |
| 2020 | Java       | 8954       | 65855.350346 |
| 2020 | C#         | 8169       | 66362.150814 |
| 2020 | TypeScript | 7904       | 69131.520622 |

Top 5 Languages for Year 2021 sorted by popularity:

| year | language   | popularity | avg_salary   |
|------|------------|------------|--------------|
| 2021 | JavaScript | 17796      | 85925.688413 |
| 2021 | Python     | 10637      | 91672.188869 |
| 2021 | TypeScript | 10131      | 88258.075313 |
| 2021 | Java       | 8114       | 87063.702859 |
| 2021 | C#         | 7811       | 84586.892075 |

Top 5 Languages for Year 2022 sorted by popularity:

| year | language   | popularity | avg_salary   |
|------|------------|------------|--------------|
| 2022 | JavaScript | 15578      | 94128.925087 |
| 2022 | TypeScript | 9892       | 96992.089466 |

|      |        |      |              |
|------|--------|------|--------------|
| 2022 | Python | 9573 | 99249.411156 |
| 2022 | C#     | 6866 | 92042.074862 |
| 2022 | Java   | 6793 | 94904.086707 |

Top 5 Languages for Year 2023 sorted by popularity:

| year | language   | popularity | avg_salary    |
|------|------------|------------|---------------|
| 2023 | JavaScript | 20912      | 105373.324455 |
| 2023 | TypeScript | 14895      | 107782.190802 |
| 2023 | Python     | 13788      | 110646.754714 |
| 2023 | C#         | 9157       | 103662.133559 |
| 2023 | Java       | 8598       | 106468.147709 |

Top 5 Languages for Year 2024 sorted by popularity:

| year | language   | popularity | avg_salary    |
|------|------------|------------|---------------|
| 2024 | JavaScript | 13587      | 98293.561861  |
| 2024 | Python     | 10077      | 102442.306738 |
| 2024 | TypeScript | 9801       | 100616.887052 |
| 2024 | C#         | 6008       | 96132.588881  |
| 2024 | Java       | 5994       | 96435.921088  |

## Calculate Pearson Correlation (Popularity vs. Salary)

For each year, compute the Pearson correlation between language popularity and average salary using Spark's correlation API.

```
In [22]: correlations = []

years = [2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024]

for y in years:
    df_year = spark.sql(f"""
        SELECT popularity, avg_salary
        FROM language_popularity_salary
        WHERE year = {y}
        """)

    if df_year.count() > 2:
        assembler = VectorAssembler(inputCols=["popularity", "avg_salary"], outputCol="features")
        df_vector = assembler.transform(df_year).select("features")
        corr_matrix = Correlation.corr(df_vector, "features", "pearson").collect()
        corr_value = corr_matrix[0, 1]
        correlations.append((y, round(float(corr_value), 3)))

print("Year | Pearson-Correlation (Popularity vs. Salary)")
for year, corr in correlations:
    print(f"{year} | {corr}")
```

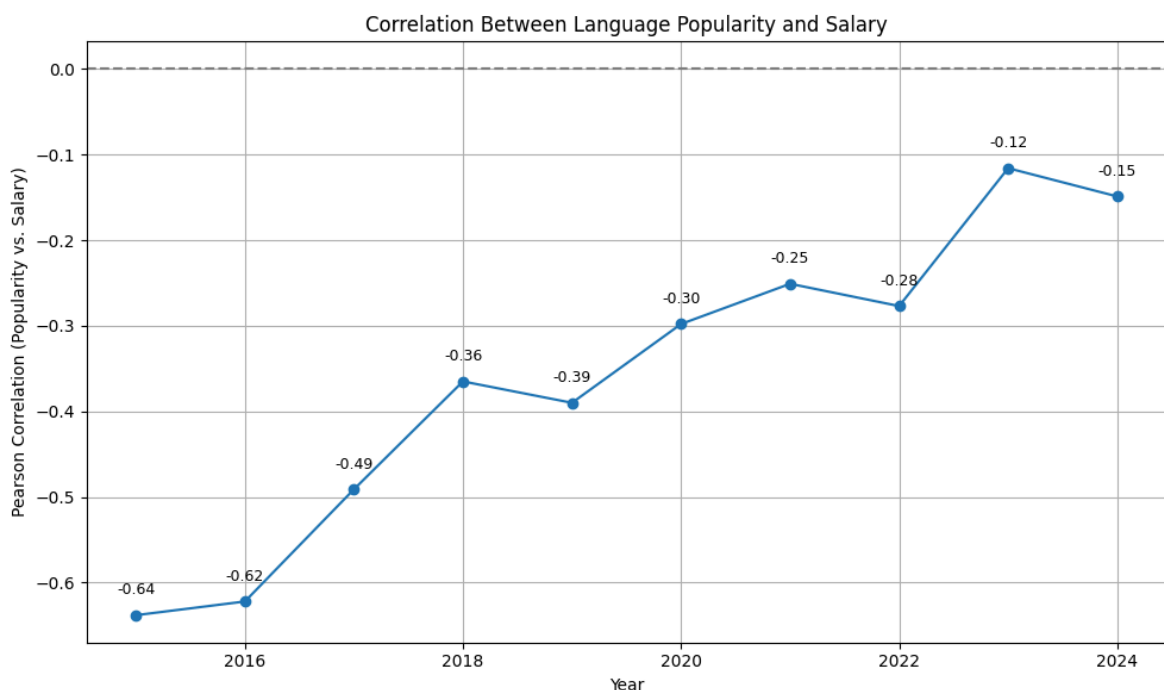
| Year | Pearson-Correlation (Popularity vs. Salary) |
|------|---------------------------------------------|
| 2015 | -0.638                                      |
| 2016 | -0.622                                      |
| 2017 | -0.491                                      |
| 2018 | -0.365                                      |
| 2019 | -0.39                                       |
| 2020 | -0.298                                      |
| 2021 | -0.251                                      |
| 2022 | -0.277                                      |
| 2023 | -0.116                                      |
| 2024 | -0.149                                      |



## Visualize Pearson Correlation Over Time

Plot a line chart to show how the correlation between language popularity and salary has changed across the years.

```
In [23]: df_corr = pd.DataFrame(correlations, columns=["year", "correlation"])
plt.figure(figsize=(10, 6))
plt.plot(df_corr["year"], df_corr["correlation"], marker="o")
plt.title("Correlation Between Language Popularity and Salary")
plt.xlabel("Year")
plt.ylabel("Pearson Correlation (Popularity vs. Salary)")
plt.axhline(0, color="gray", linestyle="--")
plt.grid(True)
plt.tight_layout()
for x, y in zip(df_corr["year"], df_corr["correlation"]):
    plt.text(x, y + 0.02, f"{y:.2f}", ha='center', va='bottom', fontsize=9)
plt.show()
```



## Calculate Spearman Correlation (Popularity vs. Salary)

For each year, compute the Spearman rank correlation to capture monotonic relationships between language popularity and salary.

```
In [24]: spearman_correlations = []

for y in years:
    df_year = spark.sql(f"""
        SELECT popularity, avg_salary
        FROM language_popularity_salary
        WHERE year = {y}
    """)

    if df_year.count() > 2:
        assembler = VectorAssembler(inputCols=["popularity", "avg_salary"], outp
```

```

df_vector = assembler.transform(df_year).select("features")
spearman_corr_matrix = Correlation.corr(df_vector, "features", method="spearmanr")
spearman_corr_value = spearman_corr_matrix[0, 1]
spearman_correlations.append((y, round(float(spearman_corr_value), 3)))

print("Year | Spearman-Correlation (Popularity vs. Salary)")
for year, corr in spearman_correlations:
    print(f"{year} | {corr}")

```

| Year | Spearman-Correlation (Popularity vs. Salary) |
|------|----------------------------------------------|
| 2015 | -0.842                                       |
| 2016 | -0.748                                       |
| 2017 | -0.715                                       |
| 2018 | -0.513                                       |
| 2019 | -0.403                                       |
| 2020 | -0.258                                       |
| 2021 | -0.255                                       |
| 2022 | -0.38                                        |
| 2023 | -0.084                                       |
| 2024 | -0.136                                       |

## Visualize Spearman Correlation Over Time

Generate a line plot to display how the Spearman correlation between language popularity and salary evolved year by year.

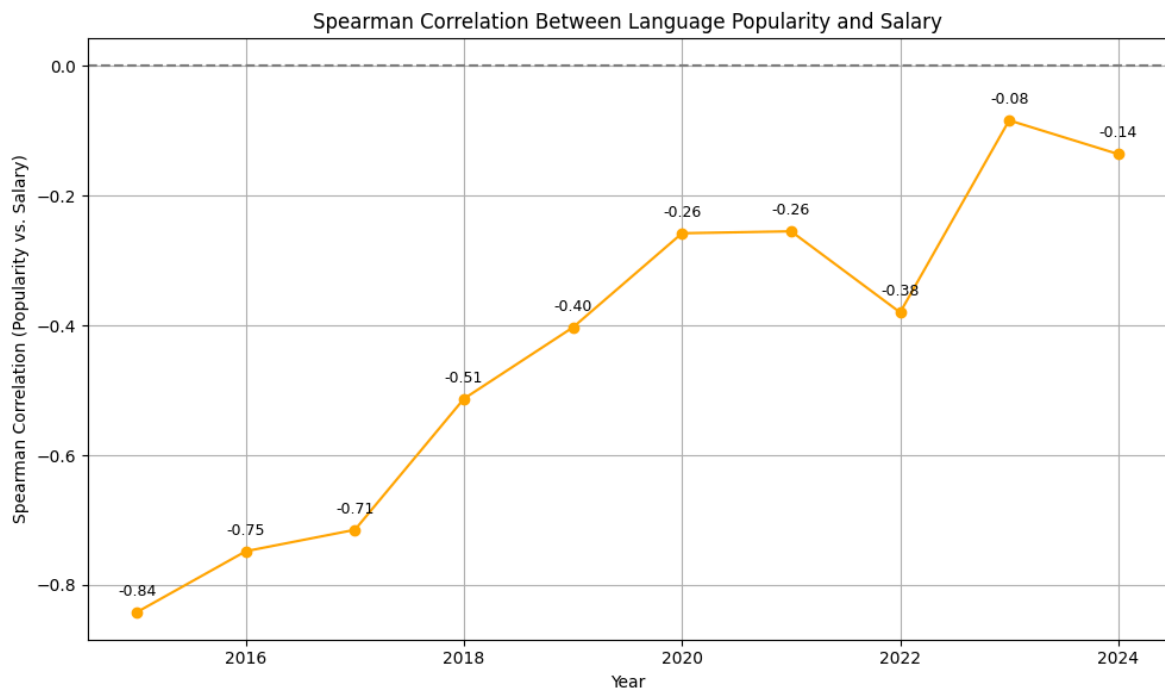
```

In [25]: df_spearman_corr = pd.DataFrame(spearman_correlations, columns=["year", "correlation"])

plt.figure(figsize=(10, 6))
plt.plot(df_spearman_corr["year"], df_spearman_corr["correlation"], marker="o",
plt.title("Spearman Correlation Between Language Popularity and Salary")
plt.xlabel("Year")
plt.ylabel("Spearman Correlation (Popularity vs. Salary)")
plt.axhline(0, color="gray", linestyle="--")
plt.grid(True)
plt.tight_layout()
for x, y in zip(df_spearman_corr["year"], df_spearman_corr["correlation"]):
    plt.text(x, y + 0.02, f"{y:.2f}", ha='center', va='bottom', fontsize=9)

plt.show()

```



## Compare Pearson and Spearman Correlation

Plot both Pearson and Spearman correlations over time to contrast linear and rank-based relationships between popularity and salary.

```
In [26]: pearson_data = [
    (2015, -0.638), (2016, -0.622), (2017, -0.491), (2018, -0.365),
    (2019, -0.390), (2020, -0.298), (2021, -0.251), (2022, -0.277),
    (2023, -0.116), (2024, -0.149)
]

spearman_data = [
    (2015, -0.842), (2016, -0.748), (2017, -0.715), (2018, -0.513),
    (2019, -0.403), (2020, -0.258), (2021, -0.255), (2022, -0.380),
    (2023, -0.084), (2024, -0.136)
]

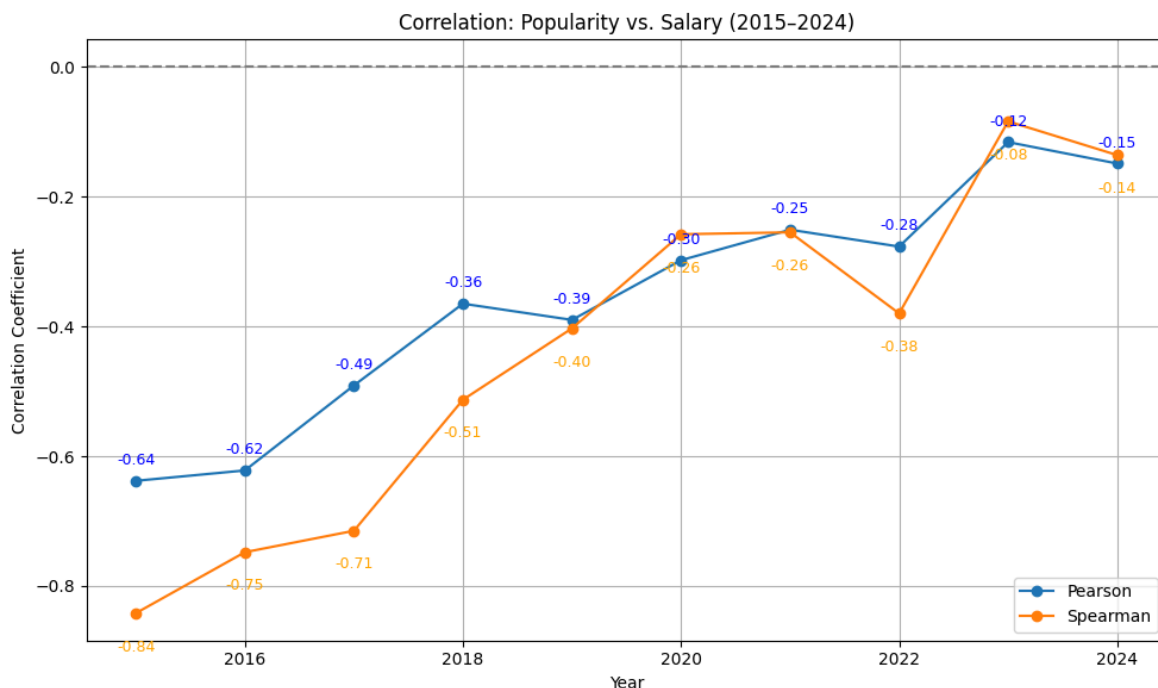
df_pearson = pd.DataFrame(pearson_data, columns=["year", "pearson"])
df_spearman = pd.DataFrame(spearman_data, columns=["year", "spearman"])

df_combined = pd.merge(df_pearson, df_spearman, on="year")

plt.figure(figsize=(10, 6))
plt.plot(df_combined["year"], df_combined["pearson"], label="Pearson", marker="o")
plt.plot(df_combined["year"], df_combined["spearman"], label="Spearman", marker="o")

plt.title("Correlation: Popularity vs. Salary (2015-2024)")
plt.xlabel("Year")
plt.ylabel("Correlation Coefficient")
plt.axhline(0, color="gray", linestyle="--")
plt.grid(True)
plt.legend()
plt.tight_layout()
for x, y in zip(df_combined["year"], df_combined["pearson"]):
    plt.text(x, y + 0.02, f"{y:.2f}", ha='center', va='bottom', fontsize=9, color='green')
for x, y in zip(df_combined["year"], df_combined["spearman"]):
    plt.text(x, y + 0.02, f"{y:.2f}", ha='center', va='bottom', fontsize=9, color='orange')
```

```
plt.text(x, y - 0.04, f"{y:.2f}", ha='center', va='top', fontsize=9, color="
plt.show()
```



## Future-Oriented Tech Career Potential Analysis

This analysis identifies the most promising combinations of programming languages, experience levels, and countries by integrating multiple weighted factors: current salary levels, popularity, long-term trend momentum, experience-based compensation, and regional earning potential. Unlike a snapshot of today's job market, this model estimates future attractiveness by combining both current and forward-looking indicators. It highlights where skills, demand, and location converge to form strong long-term opportunities.

## Language–Experience Salary View

Creates a temp view with average salary and popularity per language and experience level, excluding entries with low sample size.

```
In [27]: spark.sql("""
CREATE OR REPLACE TEMP VIEW lang_salary_exp_view AS
SELECT
    Language AS language,
    CAST(YearsCode AS INT) AS experience_years,
    AVG(ConvertedCompYearly) AS avg_salary,
    COUNT(*) AS popularity
FROM language_salary_raw
WHERE ConvertedCompYearly > 0 AND YearsCode IS NOT NULL
GROUP BY Language, CAST(YearsCode AS INT)
HAVING COUNT(*) > 50
""")
```

Out[27]: DataFrame[]

## Load Language–Experience Aggregated Salary & Popularity

```
In [28]: lang_exp_pop_df = spark.sql("SELECT * FROM lang_salary_exp_view").toPandas()
lang_exp_pop_df["experience_years"] = lang_exp_pop_df["experience_years"].astype(int)

avg_salary_map = lang_exp_pop_df.set_index(["language", "experience_years"])["avg_salary"].to_dict()
popularity_map = lang_exp_pop_df.set_index(["language", "experience_years"])["popularity"].to_dict()
```

## Load Country-Level Average Salaries

```
In [29]: df_country_salary = df_avg_salary_country.toPandas()
df_country_salary["avg_salary"] = df_country_salary["avg_salary"].round(2)
```

## Load Experience-Based Average Salaries

```
In [30]: df_exp_salary = df_avg_salary_exp_pd.copy()
df_exp_salary = df_exp_salary[df_exp_salary["YearsCode"].notna()]
df_exp_salary["YearsCode"] = df_exp_salary["YearsCode"].astype(int)
exp_salary_map = df_exp_salary.set_index("YearsCode")["avg_salary"].to_dict()
```

## Load & Calculate Trend Scores (CSV + GitHub Repo Growth)

```
In [31]: trend_csv = pd.read_csv("./RestAPIData/trend_score_repo_growth_2015_2024.csv")
trend_csv = trend_csv.dropna(subset=["trend_score"])["language", "trend_score"]
trend_csv["trend_score"] = trend_csv["trend_score"].astype(float)

df_counts = spark.sql("SELECT * FROM language_yearly_counts WHERE year BETWEEN 2015 AND 2024")
df_counts = df_counts.dropna()
df_counts["year"] = df_counts["year"].astype(int)
df_counts["repo_count"] = df_counts["repo_count"].astype(int)

trend_raw = df_counts.sort_values(["language", "year"]) \
    .groupby("language")["repo_count"].apply(lambda x: x.diff().mean()).reset_index()

scaler = MinMaxScaler()
trend_raw["trend_score"] = scaler.fit_transform(trend_raw[["trend_raw"]]) * 10
trend_raw["trend_score"] = trend_raw["trend_score"].round(1)
```

## Combine Trend Sources into Unified Score

```
In [32]: combined_trend = pd.concat([trend_csv, trend_raw[["language", "trend_score"]]])
combined_trend = combined_trend.groupby("language", as_index=False)["trend_score"].mean()
combined_trend.to_csv("./RestAPIData/combined_trend_score_2015_2024.csv", index=False)
```

## Generate All Language–Experience–Country Combinations

```
In [33]: languages = lang_exp_pop_df["language"].unique()
experience_years = sorted(exp_salary_map.keys())
countries = df_country_salary["Country"].unique()
combinations = list(product(languages, experience_years, countries))
```

## Build Feature Rows for Each Combination

```
In [34]: rows = []

for lang, exp, country in combinations:
    region_row = df_country_salary[df_country_salary["Country"] == country]
    trend_row = combined_trend[combined_trend["language"] == lang]

    avg_salary = avg_salary_map.get((lang, exp))
    popularity = popularity_map.get((lang, exp))
    exp_salary = exp_salary_map.get(exp)

    if avg_salary and popularity and not region_row.empty and exp_salary:
        rows.append({
            "language": lang,
            "experience_years": exp,
            "country": country,
            "avg_salary": avg_salary,
            "popularity": popularity,
            "exp_salary": exp_salary,
            "region_salary": region_row["avg_salary"].values[0],
            "trend_score": trend_row["trend_score"].values[0] if not trend_row.empty else 0
        })

df_combined = pd.DataFrame(rows)
```

## Normalize Features & Calculate Future Score

```
In [35]: scaler = MinMaxScaler()
df_combined[["salary_norm", "popularity_norm", "trend_norm", "exp_salary_norm", "region_norm"]] = scaler.fit_transform(df_combined[["avg_salary", "popularity", "trend_score", "exp_salary", "region_salary"]])

df_combined["future_score"] = (
    0.35 * df_combined["salary_norm"] +
    0.20 * df_combined["popularity_norm"] +
    0.15 * df_combined["trend_norm"] +
    0.20 * df_combined["exp_salary_norm"] +
    0.10 * df_combined["region_norm"]
)

df_result = df_combined.sort_values(by="future_score", ascending=False)
```

## Output: Top Results

```
In [36]: print("\nTop Language-Experience-Country Combinations for the Future:")
print(df_result[["language", "experience_years", "country",
                 "avg_salary", "exp_salary", "region_salary", "future_score"]].head(15).to_string(index=False))
```

Top Language-Experience-Country Combinations for the Future:

|    | language   | experience_years | country       | avg_salary    | exp_salary | region_salary |
|----|------------|------------------|---------------|---------------|------------|---------------|
| 62 | Python     | 28               | United States | 123679.789889 | 118203.63  | 120748.       |
| 62 | Python     | 30               | United States | 121351.976744 | 113802.77  | 120748.       |
| 62 | Python     | 25               | United States | 117644.362832 | 112049.06  | 120748.       |
| 62 | TypeScript | 28               | United States | 126072.317568 | 118203.63  | 120748.       |
| 42 | Python     | 28               | Australia     | 123679.789889 | 118203.63  | 104605.       |
| 42 | Python     | 30               | Australia     | 121351.976744 | 113802.77  | 104605.       |
| 65 | Python     | 28               | Switzerland   | 123679.789889 | 118203.63  | 103473.       |
| 65 | Python     | 30               | Switzerland   | 121351.976744 | 113802.77  | 103473.       |
| 62 | Python     | 27               | United States | 119993.593301 | 113897.92  | 120748.       |
| 62 | Python     | 24               | United States | 120489.774230 | 111133.89  | 120748.       |
| 42 | Python     | 25               | Australia     | 117644.362832 | 112049.06  | 104605.       |
| 62 | Python     | 26               | United States | 118276.776097 | 113627.98  | 120748.       |
| 57 | Python     | 28               | New Zealand   | 123679.789889 | 118203.63  | 93443.        |
| 65 | Python     | 25               | Switzerland   | 117644.362832 | 112049.06  | 103473.       |
| 57 | Python     | 30               | New Zealand   | 121351.976744 | 113802.77  | 93443.        |

```
In [37]: top_1y = df_result[df_result["experience_years"] == 1].sort_values("future_score")
top_5y = df_result[df_result["experience_years"] == 5].sort_values("future_score")
top_10y = df_result[df_result["experience_years"] == 10].sort_values("future_score")
top_all = df_result.head(1) # bereits sortiert nach future_score

summary = pd.concat([top_1y, top_5y, top_10y, top_all]).drop_duplicates()

print("\n Top Language-Experience-Country Combinations by Level:\n")
print(summary[["language", "experience_years", "country",
               "avg_salary", "exp_salary", "region_salary", "future_score"]].to_string(index=False))
```

Top Language-Experience-Country Combinations by Level:

|          | language | experience_years | country       | avg_salary    | exp_salary | region_salary |
|----------|----------|------------------|---------------|---------------|------------|---------------|
| 0.370398 | Python   | 1                | United States | 52452.161281  | 37222.96   | 120748.62     |
| 0.500582 | Python   | 5                | United States | 70246.036798  | 65683.46   | 120748.62     |
| 0.628406 | Python   | 10               | United States | 86451.392070  | 70644.65   | 120748.62     |
| 0.751759 | Python   | 28               | United States | 123679.789889 | 118203.63  | 120748.62     |

```
In [38]: spark.stop()  
         logger.info("Spark session stopped.")
```

```
2025-06-23 11:33:08,966 - INFO - Spark session stopped.
```